A SEPARATED SCHEME COVERED BY N+1 AFFINES HAS AFFINE STRATIFICATION NUMBER AT MOST N

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 21588-21593 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 21595-21605, the complete author proof (9 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Combinatorial input to the chapter's stratification-counting arguments (generic flatness stratifications used in the noetherian approximation series).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definition of affine stratification number (chapter definition section); Morphisms, lemma affine-permanence; separatedness making differences of quasi-compact opens closed (Schemes separated glueing properties).

Source reference. Original lemma 01SG: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 21605–21605. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let X be a separated scheme which has an open covering by $n+1$ affines. Then the affine stratification number of X is at most n.*

Proof. Say $X = U_0 \cup \ldots \cup U_n$ is an affine open covering. Set

$$X_i = (U_i \cup \ldots \cup U_n) \setminus (U_{i+1} \cup \ldots \cup U_n)$$

Then X_i is affine as a closed subscheme of U_i. The morphism $X_i \to X$ is affine by Morphisms, Lemma 01SG. Finally, we have $\overline{X_i} \subset X_i \cup X_{i-1} \cup \ldots X_0$. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.